Integer Quantum Resonance and Antiresonance of the Kicked Rotor: An Exact Parity Sheet

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Abstract

We solve the positive-integer resonance sheet of the quantum kicked rotor in a fixed free-after-kick convention. Parity of the integer resonance index gives a complete dichotomy: an exact Bessel propagator with ballistic momentum variance, or a state-independent two-kick antiresonance. On the resonant face we also derive the full momentum characteristic function, survival probability, and centered moments through order six.

1 Frozen Floquet owner

Let

$$\mathcal{H} = L^2(\mathbb{T}, d\theta/(2\pi)), \qquad |n\rangle(\theta) = e^{\mathrm{i}n\theta}, \qquad \widehat{n} = -\mathrm{i}\partial_\theta.$$

The periodic H^1 realization of $\widehat{n}$ is self-adjoint, and $\widehat{n}^2$ is self-adjoint on periodic H^2. For $\kappa \in \mathbb{R}$ put

$$K_\kappa = e^{-\mathrm{i}\kappa\cos\theta}, \qquad U_\tau = e^{-\mathrm{i}\tau\widehat{n}^2/2} K_\kappa.$$

Here K_κ denotes unit-modulus multiplication. Spectral calculus and bounded multiplication make both factors, hence U_τ, unitary. The ordering is part of the definition: one kick followed by free rotation.

We restrict to

$$\tau = 2\pi\ell, \qquad \ell \in \mathbb{Z}_{>0}. \tag{1}$$

No assertion below concerns another rational sheet or a detuned value.

Lemma 1 (Free-phase parity)**.** *At (1), the free factor is I for even ℓ and the half-turn*

$$(Rf)(\theta) = f(\theta + \pi)$$

for odd ℓ.

Proof. On $|n\rangle$ the free eigenvalue is $e^{-\mathrm{i}\pi\ell n^2}$. Since $n^2 - n = n(n-1)$ is even, this equals $e^{-\mathrm{i}\pi\ell n}$. It is one for even ℓ and $(-1)^n$ for odd ℓ. The latter is precisely the Fourier action of R. □

Theorem 2 (Integer resonance and antiresonance)**.** *Fix $m \in \mathbb{Z}$ and $t \in \mathbb{Z}_{\geq 0}$.*

1. *If ℓ is even, then*

$$\langle n|U_{2\pi\ell}^t|m\rangle = (-\mathrm{i})^{n-m} J_{n-m}(\kappa t). \tag{2}$$

Consequently the momentum law is $J_{n-m}(\kappa t)^2$, its mean is m, and

$$\mathrm{Var}(n) = \frac{\kappa^2 t^2}{2}, \qquad \mathbb{E}_m[\widehat{n}^2/2]_t = \frac{m^2}{2} + \frac{\kappa^2 t^2}{4}. \tag{3}$$

2. *If ℓ is odd, then $U_{2\pi\ell} = RK_\kappa$ and*

$$U_{2\pi\ell}^2 = I. \tag{4}$$

At even t the amplitude is δ_{nm}, while at odd t it is

$$(-1)^n(-\mathrm{i})^{n-m} J_{n-m}(\kappa). \tag{5}$$

Proof. For even ℓ, Lemma 1 gives $U = K_\kappa$ and $U^t = K_{t\kappa}$. The Jacobi–Anger expansion

$$e^{-\mathrm{i}x\cos\theta} = \sum_{q\in\mathbb{Z}} (-\mathrm{i})^q J_q(x) e^{\mathrm{i}q\theta}$$

with $q = n - m$ proves (2). The moment statements follow from Fourier orthogonality:

$$\sum_q J_q(x)^2 e^{\mathrm{i}qu} = J_0(2x\sin(u/2)).$$

Two derivatives at $u = 0$ give mean displacement zero and variance $x^2/2$; translation by m gives (3).

For odd ℓ, the free factor is R. Since the half-turn reverses the cosine,

$$RK_\kappa R = K_{-\kappa} = K_\kappa^{-1}.$$

Thus $(RK_\kappa)^2 = I$. Even powers are I, and odd powers are RK_κ. Applying R after the Jacobi–Anger expansion multiplies the nth Fourier coefficient by $(-1)^n$, which proves (5). ☐

The phrase *operator parity owner* records the first paper round: the all-time dichotomy is an operator theorem, not a finite-lattice simulation.

2 Exact momentum statistics

Let $x = \kappa t$ on the even sheet and set $q = n - m$. If $c_q = (-\mathrm{i})^q J_q(x)$, Fourier orthogonality gives

$$\begin{aligned}\Phi_x(u) &= \sum_{q\in\mathbb{Z}} |c_q|^2 e^{\mathrm{i}qu} \\ &= \frac{1}{2\pi}\int_0^{2\pi} e^{-\mathrm{i}x\cos(\theta+u)} e^{\mathrm{i}x\cos\theta}\, d\theta = J_0(2x\sin(u/2)).\end{aligned} \tag{6}$$

The last equality follows by shifting the integration variable in the standard integral representation of J_0; its sign is immaterial because J_0 is even. At $u = 0$, (6) also proves normalization.

Theorem 3 (Characteristic and moment closure)**.** *For even ℓ, the survival probability is*

$$\Pr_m\{n_t = m\} = J_0(\kappa t)^2,$$

and the centered odd moments through order six vanish. The nonzero centered moments are

$$\begin{aligned}\mu_2 &= \frac{x^2}{2}, \\ \mu_4 &= \frac{x^2}{2} + \frac{3x^4}{8}, \\ \mu_6 &= \frac{x^2}{2} + \frac{15x^4}{8} + \frac{5x^6}{16}.\end{aligned} \tag{7}$$

For odd ℓ, the law is the point mass at m at even times and the one-kick Bessel law $J_{n-m}(\kappa)^2$ at odd times; use $x = 0$ and $x = \kappa$, respectively, in (7)*.*

Proof. The survival formula is (2) at $n = m$. For a characteristic function, $\mathbb{E}(q^r) = \Phi_x^{(r)}(0)/\mathrm{i}^r$. Six direct differentiations of the final expression in (6) give (7); evenness in u annihilates the odd derivatives. The odd-sheet statement follows from Theorem 2. ☐

The phrase *Bessel characteristic owner* marks the substantive first revision. It upgrades a variance statement to an exact distributional identity and moments through sixth order.

References

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